

Finite-Domain Quantified Nullivance Logic: A Machine-Checked Completeness Theorem for a Macro-Free Extensional Tableau*

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Abstract

We extend nullivance propositional logic (NPL) — a paraconsistent, two-channel logic on the unit square with a harmonization connective $\oplus$ and four threshold signs T^+, T^-, F^+, F^- — by finite-domain quantifiers. The language is function-free first-order with crisp equality; models interpret predicates over a fixed finite nonempty domain by values in the four-element matrix **4**, and the finite universal (existential) quantifier takes the pointwise minimum (maximum) on the truth channel and the dual on the falsity channel. Main results, each machine-checked in Lean 4: (i) the exact threshold projection of the propositional theory lifts to the quantified language over finite domains; (ii) three tempting proof routes are *refuted* by explicit countermodels: the naive lifted tableau (crisp $x=x$ is valid but underivable), the bare grounding reduction to propositional consequence, and a dispatcher shortcut in the simulation program; (iii) the repaired *core extensional tableau*, whose closure conditions add four crisp-equality clauses and two ground-closure clauses to the decomposition rules, is sound and *complete*: a branch closes exactly when no finite model satisfies it. The completeness proof is a quantified tableau engine driven by a domain-weighted measure ($\text{qsize}(\forall x \varphi) = (n+1) \cdot \text{qsize}(\varphi) + 1$ over a domain of size $n+1$) under which every rule strictly decreases branch

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weight, together with a canonical-model construction at the literal stage keyed by an injective ground-atom encoding. As a corollary, closure of the grounded propositional branch under rigid constraints reaches the core calculus with no propositional-simulation macro — settling the artifact’s final open bridge conjecture. We also document the failure analysis that led from a derivation-by-derivation simulation program to the weighted-measure proof.

1 Introduction

The companion paper develops the propositional core of nullivance logic (NPL): formulas valued in $[0, 1]^2$ with independent truth and falsity channels, a threshold $\tau \in (0, 1]$ inducing four signs, an exact projection onto the four-valued matrix **4**, and a four-signed analytic tableau calculus with a machine-checked completeness theorem. This paper asks the next question: what does a *quantified* NPL look like, and can its metatheory be machine-checked to the same standard?

We deliberately restrict to **finite nonempty domains**. The restriction is not cosmetic: the exact-projection theorem — the load-bearing bridge between the continuous semantics and the finite matrix — uses that finite infima and suprema are attained, and it can fail over infinite domains when a supremum is approached but not attained (Remark 3.7). Finite domains keep the entire two-layer architecture of NPL intact, and they are the natural first-order setting for the intended applications of a threshold logic (finite data, finite scenarios). Equality is *crisp*: $x=y$ evaluates to **T** or **F** by identity of domain elements, never to a glut or a gap.

Contributions. All results are verified sorry-free in Lean 4 unless explicitly marked otherwise; [L] markers name the verifying declarations.

1. *Finite-domain quantified NPL* (Section 3): function-free syntax with predicate atoms, crisp equality, the NPL connectives including $\oplus$, and quantifiers evaluated by finite channel-wise extrema; quantifier duality under channel-swap negation; and the exact threshold projection for the quantified language.
2. *A catalogue of machine-checked refutations* (Section 5): the naive signed tableau is incomplete (crisp reflexivity is valid but underivable); the bare grounding bridge to propositional NPL is unsound as a completeness route; and a natural dispatcher shortcut for the completeness proof

fails on a two-element counterexample. Each failure shaped the final calculus.

3. *The core extensional tableau and its exact semantic characterization* (Sections 6 and 7): the headline theorem states that a finite branch closes in the macro-free core calculus *iff* it has no finite model, hence derivability coincides with finite semantic consequence. The proof introduces a domain-weighted formula measure under which every rule of the calculus — including the $(n+1)$ -ary quantifier rules — strictly decreases the branch weight, and closes the literal stage by a canonical model read off the branch through an injective ground-atom encoding.
4. *The constrained-grounding corollary* (Section 8): closure of the grounded propositional branch augmented by *rigid constraints* (fixing the identity and equality atoms) yields core closure. Unlike the first completeness theorem of the artifact, no propositional-simulation macro rule appears anywhere: the propositional derivation is consumed through its soundness and the core closure is rebuilt semantically.
5. *A documented dead end* (Section 9): the derivation-by-derivation simulation program (replay traces with residual fold tails) is developed to the point where its exact obstruction is isolated — the binary propositional decomposition cascade cannot be reassembled into the core’s $(n+1)$ -ary quantifier rules — and the weighted-measure proof is shown to dissolve the obstruction. The simulation layer remains machine-verified as a study of the approach.

Isn’t this just propositional logic? A reader may object that finite-domain many-valued logic reduces routinely to propositional logic: ground out the quantifiers as finite folds and apply the companion paper’s completeness theorem. Three machine-checked facts say the reduction is not routine. First, the naive lifted tableau is incomplete — crisp reflexivity is valid and undervivable (Proposition 5.1). Second, the *bare* grounding bridge is refuted: propositional consequence of the grounded branch does not track finite-domain consequence, because propositional valuations may assign the identity and equality atoms values no finite-domain model induces (Proposition 5.3); repairing this needs the rigid constraints of Definition 4.3. Third, even with rigid constraints, the repaired route initially proved completeness only for a calculus with a propositional *simulation macro* — a single rule quoting the entire propositional tableau — which a proof-theoretic reading wants eliminated; eliminating it is exactly the content of Theorems 7.1 and 8.1, and

the documented failure of the direct macro-elimination attempt (Section 9) shows the elimination has real content.

Method. The development follows the project’s rigor rules: every claim is labeled, refutation is attempted before proof, and the Lean artifact is the ground truth. The manuscript numbering is presentational; the stable numbering and the full proofs live in the repository documentation.

2 Preliminaries: propositional NPL

We summarize what is needed from the companion paper. The propositional language has atoms and $\neg, \wedge, \vee, \oplus$. The matrix **4** is $\{0, 1\}^2$ with corners **T** = (1, 0), **F** = (0, 1), **B** = (1, 1), **N** = (0, 0); negation swaps the two bits, $\wedge$ acts as (min, max), $\vee$ as (max, min), and $\oplus$ as (min, min) (bilattice consensus [8, 7, 1]). The $\{\neg, \wedge, \vee\}$ -fragment is Belnap–Dunn first-degree entailment [2, 6]. Four signs read the bits: T^+ (t -bit is 1), T^- (t -bit is 0), F^+ , F^- likewise on the f -bit. A four-signed analytic tableau calculus (sixteen decomposition rules, two closure conditions) is sound and complete for signed consequence over **4**, and — by the exact projection — over the continuous semantics on $[0, 1]^2$; signed tableaux over sets-of-values signs are a standard paradigm [9]. All of this is machine-checked in the same artifact this paper extends.

3 Finite-domain quantified NPL

Definition 3.1 (Syntax). Fix countable stocks of variables and predicate symbols. Quantified formulas are generated by

$$\varphi ::= P(x_0, \dots, x_k) \mid x = y \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \varphi \oplus \varphi \mid \forall x \varphi \mid \exists x \varphi.$$

Predicate atoms carry a list of variables; arity is a well-formedness discipline of the intended signature, and evaluation is total on the raw syntax. [L:FiniteF0.QFormula]

Definition 3.2 (Models and assignments). For a fixed $n \in \mathbb{N}$, the domain is $\text{Fin}(n+1)$ — finite and nonempty by construction. A *model* M assigns to each predicate symbol and argument list a value in **4**. An *assignment* ρ maps variables to domain elements; $\rho[x \mapsto d]$ updates it. [L:FiniteF0.QModel, FiniteF0.Assignment]

Definition 3.3 (Evaluation). Evaluation extends the propositional **4** clauses by crisp equality and finite channel-wise quantifiers:

$$V(x=y) = \begin{cases} \mathbf{T} & \rho(x) = \rho(y) \\ \mathbf{F} & \text{otherwise,} \end{cases} \quad V(\forall x \varphi) = \left(\bigwedge_d t_d, \bigvee_d f_d \right), \quad V(\exists x \varphi) = \left(\bigvee_d t_d, \bigwedge_d f_d \right),$$

where (t_d, f_d) is the value of φ under $\rho[x \mapsto d]$ and the extrema range over the finite domain. Signed satisfaction reads one bit of the value, exactly as in the propositional case. [L:FiniteF0.qeval, forallV4, existsV4]

Definition 3.4 (Branches; semantic consequence). A *signed branch member* is a triple $s(\rho, \varphi)$ of a sign, an assignment, and a formula — each member carries its *own* assignment. A *branch* is a finite list of members; a model *satisfies* a branch iff it satisfies every member. *Semantic consequence* is domain-indexed: $\Gamma \models_{Q,n} s\varphi$ iff every model over $\text{Fin}(n+1)$ that satisfies every member of Γ satisfies $s\varphi$. [L:FiniteF0.QSigned, qsatSigned, qsatBranch, QConsequence4]

Proposition 3.5 (Quantifier duality). $\neg \forall x \varphi$ and $\exists x \neg \varphi$ have the same value, as do $\neg \exists x \varphi$ and $\forall x \neg \varphi$: channel-swap negation exchanges the two finite extrema. The nonempty domain is used; over an empty domain the statement would depend on conventions for vacuous quantifiers. [L:FiniteF0.qeval_neg_all, qeval_neg_ex]

Theorem 3.6 (Finite exact projection). Interpret predicates by continuous truth-objects in $[0,1]^2$ and evaluate quantifiers by finite channel-wise infima/suprema. For every threshold $0 < \tau \leq 1$, thresholding commutes with evaluation: $\pi_\tau(V_M(\varphi)) = V_{\pi_\tau(M)}(\varphi)$, where $\pi_\tau(M)$ projects the predicate interpretations pointwise. Hence the finite-domain metatheory, like the propositional one, can be developed on **4** and lifted. [L:FiniteF0.finite_exact_projection]

Remark 3.7 (Why finite domains). Over an infinite domain the existential/supremum direction of Theorem 3.6 can fail when the supremum is not attained: every instance can fall below the threshold while the supremum reaches it. The finite, nonempty domain is therefore a load-bearing design choice, not a convenience. (Recorded as a refutation attempt in the development.)

4 Grounding

The finite domain makes every quantifier a finite fold, which suggests translating quantified branches into propositional ones.

Definition 4.1 (Ground atoms and the ground translation). *Ground atoms* are: two fixed identity atoms $\top_g$ and $\perp_g$ (fold units), a ground predicate atom for each predicate symbol and argument list, and a ground equality atom for each pair of domain elements. Ground atoms are encoded injectively into propositional atoms. The translation gr_ρ maps a quantified formula under an assignment to a propositional formula: predicate and equality atoms to their ground atoms; connectives homomorphically; $\forall x \varphi$ to the finite conjunction fold of $\{\text{gr}_{\rho[x \mapsto d]}(\varphi)\}_d$ (with unit $\top_g$) and $\exists$ to the disjunction fold (with unit $\perp_g$). [L:FiniteFO.GroundAtom, groundAtomCode, FiniteFO.ground, foldConj, foldDisj]

Lemma 4.2 (Grounding truth lemma). *For every model M there is a propositional valuation v_M (reading ground atoms through M , with $\top_g \mapsto \mathbf{T}$, $\perp_g \mapsto \mathbf{F}$, and equality atoms crisp) such that the propositional value of $\text{gr}_\rho(\varphi)$ under v_M equals the quantified value of φ under M, ρ . [L:FiniteFO.ground_truth, groundVal]*

Definition 4.3 (Rigid constraints). The *rigid constraints* form the finite propositional branch containing T^+, F^- for $\top_g$; T^-, F^+ for $\perp_g$; and for every pair (a, b) of domain elements, T^+, F^- (respectively T^-, F^+) for the ground equality atom of (a, b) when $a = b$ (respectively $a \neq b$). They pin exactly the part of a ground valuation that finite-domain models do not choose freely; ground predicate atoms remain unconstrained. [L:FiniteFO.rigidGroundConstraints]

5 Calculi, and what fails

The propositional decomposition rules lift to signed quantified branch members, acting on the member’s own assignment. For self-containedness we reproduce the sixteen propositional rows (from the companion paper; “ A, B ” adds both members to the branch, “ $A \mid B$ ” splits it):

	T^+	T^-	F^+	F^-
$\neg\varphi$	$F^+\varphi$	$F^-\varphi$	$T^+\varphi$	$T^-\varphi$
$\varphi \wedge \psi$	$T^+\varphi, T^+\psi$	$T^-\varphi \mid T^-\psi$	$F^+\varphi \mid F^+\psi$	$F^-\varphi, F^-\psi$
$\varphi \vee \psi$	$T^+\varphi \mid T^+\psi$	$T^-\varphi, T^-\psi$	$F^+\varphi, F^+\psi$	$F^-\varphi \mid F^-\psi$
$\varphi \oplus \psi$	$T^+\varphi, T^+\psi$	$T^-\varphi \mid T^-\psi$	$F^+\varphi, F^+\psi$	$F^-\varphi \mid F^-\psi$

The eight finite quantifier rules come in two shapes, determined by which channel bit the sign reads of the finite extrema:

<i>block</i> (add all $n+1$ instances, non-branching)	$T^+\forall, F^-\forall, T^-\exists, F^+\exists$
<i>per-element</i> (one closed child per domain element)	$T^-\forall, F^+\forall, T^+\exists, F^-\exists$

Closure, in the naive calculus, is an identical opposite-signed pair (same assignment, same formula). [L:FiniteFO.QCloses]

Proposition 5.1 (The naive calculus is incomplete). *Crisp reflexivity $T^+(x=x)$ is valid over every finite domain, but the empty branch does not derive it: no decomposition rule applies to an equality atom and no closure pair exists. Machine-checked countermodel to the completeness conjecture for QCloses. [L:FiniteFO.qeqRefl0_valid, qeqRefl0_not_derivable, qcompleteness_current_refuted]*

Definition 5.2 (Equality-completed calculus). Add four crisp-equality closure clauses: a branch closes if it contains $T^-(x=y)$ or $F^+(x=y)$ under an assignment with $\rho(x) = \rho(y)$, or $T^+(x=y)$ or $F^-(x=y)$ with $\rho(x) \neq \rho(y)$. [L:FiniteFO.QClosesEq]

Proposition 5.3 (The bare grounding bridge fails). *Grounding alone does not reduce finite-domain consequence to propositional consequence: $T^+(x=x)$ is a finite-domain validity, but the grounded sequent is not a propositional consequence — a propositional valuation may assign the ground equality atom of $(0,0)$ any value, while every finite-domain model forces it to **T**. Machine-checked countermodel; the rigid constraints of Definition 4.3 repair exactly this gap. [L:FiniteFO.qeqRefl0_ground_not_consequence4, qeqRefl0_ground_branch_not_closes]*

Example 5.4 (Why equality clauses are not enough). Let $\rho(x) = \sigma(y) = d$ and consider the two-member branch $\{T^+(\rho, Px), T^-(\sigma, Py)\}$. Every model gives both members the same ground predicate value at $(P, [d])$, so the branch is unsatisfiable; but the members are not an identical pair (different formulas, different assignments), no rule decomposes literals, and no equality clause applies. Closure must be allowed on *equal groundings*, not merely on identical members.

6 The core extensional tableau

Definition 6.1 (Core calculus). The *core extensional finite-domain tableau* extends the equality-completed calculus of Definition 5.2 by two *ground closure clauses*: a branch closes if it contains members $T^+(\rho, \varphi)$ and $T^-(\sigma, \psi)$

with $\text{gr}_\rho(\varphi) = \text{gr}_\sigma(\psi)$, or F^+/F^- likewise. All sixteen propositional decomposition rules and all eight finite quantifier rules are retained. The calculus contains *no* propositional simulation macro. Derivability: $\Gamma \vdash_{\text{core}} s\varphi$ iff the branch $\{\bar{s}\varphi\} \cup \Gamma$ closes, where $\bar{s}$ is the opposite sign. [L:FiniteF0.QClosesExtCore, QDerivesExtCore]

Proposition 6.2 (Soundness). *If a branch closes in the core calculus, no finite model satisfies it; hence $\Gamma \vdash_{\text{core}} s\varphi$ implies $\Gamma \models_{Q,n} s\varphi$.* [L:FiniteF0.QClosesExtCore.unsat]

Proof sketch. Induction on the closure derivation. Each closure condition is unsatisfiable at its own members: an identical pair by sign complementarity; an equality clause by crispness of $=$; a ground-closure pair because Lemma 4.2 gives both members the same value under any model, and opposite signs of one value cannot both hold. Each decomposition rule preserves satisfiability downward into (one of) its children by the corresponding local satisfaction equivalence — for the quantifier rules, the signed reading of the finite extrema. $\square$

7 The completeness theorem

Theorem 7.1 (Semantic completeness of the core calculus). *Let B be a finite branch of signed quantified members over domain size $n+1$. If no model with domain $\text{Fin}(n+1)$ satisfies B , then B closes in the core calculus. With Proposition 6.2, closure is exactly finite unsatisfiability, and*

$$\Gamma \vdash_{\text{core}} s\varphi \iff \Gamma \models_{Q,n} s\varphi.$$

[L:FiniteF0.QClosesExtCore.complete_of_unsat, qclosesExtCore_iff_unsat, QDerivesExtCore.complete, qDerivesExtCore_iff_qconsequence4]

Proof (engine; full details in the artifact). Step 1: the domain-weighted measure. Define qsize on quantified formulas by $\text{qsize} = 1$ on predicate and equality atoms, $+1$ for negation, $+1$ over the sum for binary connectives, and

$$\text{qsize}(\forall x \varphi) = \text{qsize}(\exists x \varphi) = (n+1) \cdot \text{qsize}(\varphi) + 1,$$

extended to branches by summation. Every rule of Definition 6.1 strictly decreases the weight of the undecomposed segment: unary and binary rules drop at least the head's outermost symbol; a block quantifier rule replaces weight $(n+1)w + 1$ by the $n+1$ instances of weight w each; a per-element

rule replaces it by a single instance of weight $w \leq (n+1)w$. The case analysis is exhaustive over the eight formula constructors and four signs.

Step 2: the engine. By strong induction on the weight bound, a branch split as *todo* + *lits* (with *lits* literals) that no finite model satisfies closes in the core: a literal head moves to *lits*; a compound head is decomposed by its rule, unsatisfiability transfers to each child by the corresponding local satisfaction equivalence (the signed quantifier equivalences and the propositional local-soundness table), the induction hypothesis applies at strictly smaller weight, and monotonicity of closure under branch extension [L: QClosesExtCore.mono] restores the full branch. Because the core’s quantifier rules are natively $(n+1)$ -ary, no intermediate “partially decomposed fold” state ever arises (contrast Section 9).

Step 3: the literal stage. For a branch of literals: if an equality literal contradicts its crisp value, an equality clause closes; if two literals carry T^+/T^- (or F^+/F^-) with equal groundings, a ground closure clause closes. Otherwise build the *canonical model*: the truth (falsity) bit of each ground predicate atom is set iff some T^+ - (F^+ -) literal grounds to it, read through the injective ground-atom encoding. Positive literals are satisfied by their own witness; negative literals because the opposite witness would be a ground closure pair, excluded by assumption; equality literals by crispness, their four unsatisfiable sign/value combinations being exactly the equality clauses. This contradicts unsatisfiability, so a closure option applied. $\square$

Remark 7.2 (What the refutations bought). Three hypotheses of Theorem 7.1 carry machine-checked failure witnesses for their removal: without the ground closure clauses, Example 5.4 is unsatisfiable but open; without the $(n+1)$ -weighting, decomposing a block quantifier *increases* the unweighted measure and the induction has no footing; without any one of the four equality clauses, the corresponding literal configuration is unsatisfiable but unmodelable and unclosable.

8 Constrained grounding reaches the core

The artifact’s first finite-domain completeness theorem ran through the propositional calculus: if the grounded branch together with the rigid constraints closes propositionally, an *extensional macro calculus* closes the quantified branch by a single simulation rule. That macro rule is exactly what a proof-theoretic reading wants to eliminate.

Theorem 8.1 (Constrained grounding, macro-free). *For every finite branch B : if the propositional branch $\text{Rigid}_n + \text{gr}(B)$ closes in the propositional tableau calculus, then B closes in the core calculus of Definition 6.1. No simulation macro occurs: the propositional derivation is consumed through propositional soundness, and the core closure is rebuilt by Theorem 7.1. Concretely: a finite model of B would induce, by Lemma 4.2 and the rigid values, a propositional valuation satisfying $\text{Rigid}_n + \text{gr}(B)$, contradicting propositional soundness; hence B is finitely unsatisfiable and Theorem 7.1 applies. [L:FiniteFO.groundBranch_closes_to_core, satBranch_groundVal_rigid]*

Conversely, grounding is sound: the grounded branch of a satisfiable quantified branch is satisfiable under the induced valuation (Lemma 4.2), so Theorem 8.1 composes with the propositional completeness theorem of the companion paper into a round-trip between the two levels.

9 The simulation program: a documented dead end

An earlier phase of the project attempted Theorem 8.1 by *simulation*: induction on the propositional closure derivation, maintaining a replay trace whose items include quantified members, rigid atoms, and *residual fold tails* — the partially decomposed remains of a grounded quantifier, which are not groundings of any single quantified formula. The layer produced a complete, machine-verified dispatch analysis for propositional closure pairs (sixteen source combinations per closure sign), a recursive “tail consumer” closing any quantified formula against a fold tail it grounds into, and two further machine-checked refutations of shortcut routes.

The program then hit a precise wall: when the propositional derivation decomposes the grounding of a *per-element-sign* quantifier ($T^-\forall$ and its mirrors), the binary conjunction cascade produces a right child asserting the pooled fold of the remaining instances in one branch, while the core rule needs one closed child per domain element; the pooled induction hypothesis cannot be redistributed, and depth-preserving inversion does not make the induction well-founded. The weighted-measure engine of Theorem 7.1 dissolves the problem at the root: the core’s quantifier rules are $(n+1)$ -ary, so the binary cascade never appears. We record the dead end because the failure analysis, not luck, produced the measure; the simulation layer remains in the artifact as a verified study.

10 Related work

Finitely-valued first-order tableaux. Complete tableau calculi for arbitrary finite many-valued first-order logics are classical: Carnielli’s systematization [5] proves abstract completeness, model existence, compactness and Löwenheim–Skolem theorems for tableaux with *distribution quantifiers*, and Hähnle’s monograph [9] generalizes the sign language to sets-as-signs. We therefore claim no novelty for many-valued first-order tableau completeness as such, and NPL’s four threshold signs are themselves sets-as-signs instances ($\{\mathbf{T}, \mathbf{B}\}$, $\{\mathbf{F}, \mathbf{N}\}$, $\{\mathbf{B}, \mathbf{F}\}$, $\{\mathbf{T}, \mathbf{N}\}$). The present calculus differs in format and scope: quantifiers range over a *fixed finite nonempty domain* and are decomposed by $(n+1)$ -ary instance rules rather than through value distributions over arbitrary domains; the closure conditions include crisp-equality clauses and ground-closure (extensional) clauses, neither of which appears in the cited calculi, whose languages omit equality; and the completeness proof is driven by the domain-weighted measure, which has no counterpart in the distribution-quantifier setting. Two delimitations are worth making explicit. Semantically, fixing the domain *in the judgment* is a design choice, not a lack: the intended applications quantify over known finite data, and the domain-indexed consequence relation (Definition 3.4) is the natural primitive there — the distribution-quantifier tradition answers a different question (consequence over all structures). Proof-theoretically, one may ask whether the core calculus is an instance of the sets-as-signs framework whose generic completeness argument applies off the shelf: the decomposition rules are, but the crisp-equality clauses and the ground-closure clauses act on *pairs of members across different assignments* through the grounding translation, which is outside the per-formula format of the generic argument; Example 5.4 is precisely a branch that every sets-as-signs closure notion keyed to identical signed formulas leaves open.

Quantified Belnap–Dunn logics. First-order FDE and its tableaux are textbook material [10], on the classical foundations of [2, 6]. Recent work studies free quantification over four-valued and fuzzy bilattice-valued logics with dual-domain semantics for non-denoting terms [4]; that axis (free logic, unrestricted domains, no mechanized proof theory) is orthogonal to ours (total denotation, fixed finite domains, machine-checked exact characterization). The harmonization connective $\oplus$ in the object language and the threshold reading remain the NPL-specific propositional base, positioned against the bilattice literature in the companion paper.

Machine-checked completeness. Formalized completeness theorems exist for classical first-order logic — sequent calculi [11] and a coinductive frame-

work with a formalized tableau for many-sorted classical logic with equality [3] — and for modal, hybrid and intuitionistic systems. Our literature pass (recorded in the repository, `references/npl-positioning.md` §6, 2026-07-09) found no machine-checked completeness theorem for a many-valued quantified tableau calculus; to our knowledge, the present development is the first, and we phrase this as a search-scoped claim rather than an absolute one.

11 Open problems

(1) *Infinite domains*: the projection argument fails at unattained suprema (Remark 3.7); a quantified NPL over infinite domains needs either attained-extrema semantics or a genuinely different bridge. (2) *Function symbols and non-crisp equality*: the present language is function-free and its equality never takes glut or gap values; both restrictions are design choices worth revisiting. (3) *Complexity*: finite-domain consequence is decidable by finiteness of the model space; its exact complexity as a function of domain size and formula size is open here. (4) *The trace-calculus bridge*: the optional sharpening that the replay-trace calculus itself absorbs constrained propositional closure remains a conjecture in the repository (no longer on any critical path).

A The Lean artifact

The results of this paper are verified in the Nullivance Lean 4 library (module `Nullivance.FiniteF0`, ≈ 380 declarations), part of the same artifact as the companion paper. **Availability.** The library will be archived with a DOI at submission (the repository carries Zenodo metadata); until then it is available from the author on request. **Toolchain.** Lean 4 (`leanprover/lean4:v4.32.0-rc1`) with `mathlib` pinned via the Lake manifest; `lake exe cache get` && `lake build` (approx. 2000 jobs). **Coverage policy.** The build is sorry-free; every result above is stated against the calculus `QClosesExtCore` directly. Axiom audit: every named declaration depends only on `propext`, `Classical.choice`, `Quot.sound`. **Key declarations.** Semantics: `qeval`, `qsatBranch`; projection: `finite_exact_projection`; refutations: `qcompleteness_current_refuted`, `qeqRef10_not_derivable`; calculus: `QClosesExtCore`; soundness: `QClosesExtCore.unsat`; completeness engine: `qsize`, `qweightB`, `qclosesCore_todo`, `qclosesCore_lits`; headline: `QClosesExtCore.complete_of_unsat`, `qDerivesExtCore_iff_qconsequence4`;

grounding: `ground_truth`, `groundBranch_closes_to_core`.

References

- [1] Ofer Arieli and Arnon Avron. Reasoning with logical bilattices. *Journal of Logic, Language and Information*, 5(1):25–63, 1996.
- [2] Jr. Belnap, Nuel D. A useful four-valued logic. In J. Michael Dunn and George Epstein, editors, *Modern Uses of Multiple-Valued Logic*, pages 5–37. D. Reidel, Dordrecht, 1977. Page range per CrossRef record of the chapter DOI.
- [3] Jasmin Christian Blanchette, Andrei Popescu, and Dmitriy Traytel. Soundness and completeness proofs by coinductive methods. *Journal of Automated Reasoning*, 58(1):149–179, 2017.
- [4] Libor Běhounek, Martina Daňková, and Antonín Dvořák. Free quantification in four-valued and fuzzy bilattice-valued logics, 2023. Presented at IUKM 2023; arXiv:2306.13079.
- [5] Walter A. Carnielli. Systematization of finite many-valued logics through the method of tableaux. *The Journal of Symbolic Logic*, 52(2):473–493, 1987.
- [6] J. Michael Dunn. Intuitive semantics for first-degree entailments and “coupled trees”. *Philosophical Studies*, 29(3):149–168, 1976.
- [7] Melvin Fitting. Bilattices and the semantics of logic programming. *Journal of Logic Programming*, 11(2):91–116, 1991.
- [8] Matthew L. Ginsberg. Multivalued logics: A uniform approach to reasoning in artificial intelligence. *Computational Intelligence*, 4(3):265–316, 1988.
- [9] Reiner Hähnle. *Automated Deduction in Multiple-Valued Logics*, volume 10 of *International Series of Monographs on Computer Science*. Oxford University Press, Oxford, 1994. Sets-as-signs signed tableaux for finitely-valued logics.
- [10] Graham Priest. *An Introduction to Non-Classical Logic: From If to Is*. Cambridge University Press, Cambridge, 2 edition, 2008. Ch. 8: FDE, incl. signed tableaux with nodes $A, + / A, -$.

- [11] Jørgen Villadsen and Anders Schlichtkrull. Formalizing a paraconsistent logic in the Isabelle proof assistant. In *Transactions on Large-Scale Data- and Knowledge-Centered Systems XXXIV*, Lecture Notes in Computer Science, pages 92–122. Springer, 2017.